



Walailak University

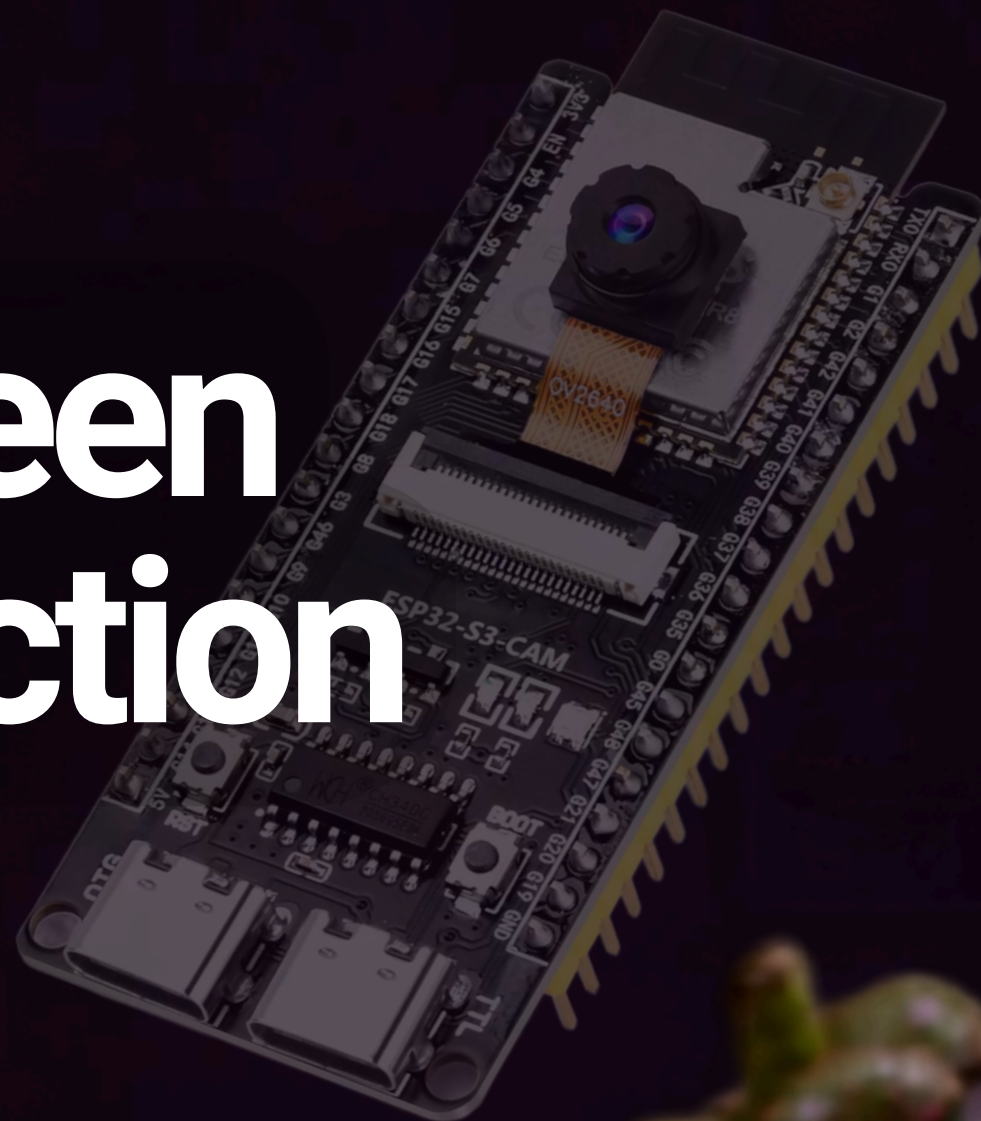
AIoT Mangosteen Ripeness Detection

AI and Computer Vision Technology for
Accurate Quality Control, Reduced Loss, Increased Value

Presented to

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COE67-312 AI in Embedded Systems



Problems and Motivation



Problem of Sorting Mangosteens Using Manual Labor or Visual Inspection

Limited overall dataset size
Imbalanced classes, especially the Overripe class



Motivation for Development

Manual ripeness inspection may be inconsistent and time-consuming.
Edge AI can provide fast and automatic mangosteen ripeness detection directly on a low-cost device.

Objective

Develop an ESP32-S3 AIoT system to classify mangosteen as Unripe, Ripe, or Overripe.

TARGET : >90% overall test accuracy



Dataset Overview

237

original images

20

ambiguous images excluded

217

images used for train / validation / test

Split	Unripe	Ripe	Overripe	Total
Train	74	58	19	151
Val	15	11	4	30
Test	20	12	4	36
Total	109	81	27	217

Data handling

- Review labels before splitting
- Split original images first
- Apply rotation / flip only to training data
- Keep excluded images outside evaluation

Unripe



Ripe



Overripe



Model Architecture

Dual-Feature Extraction

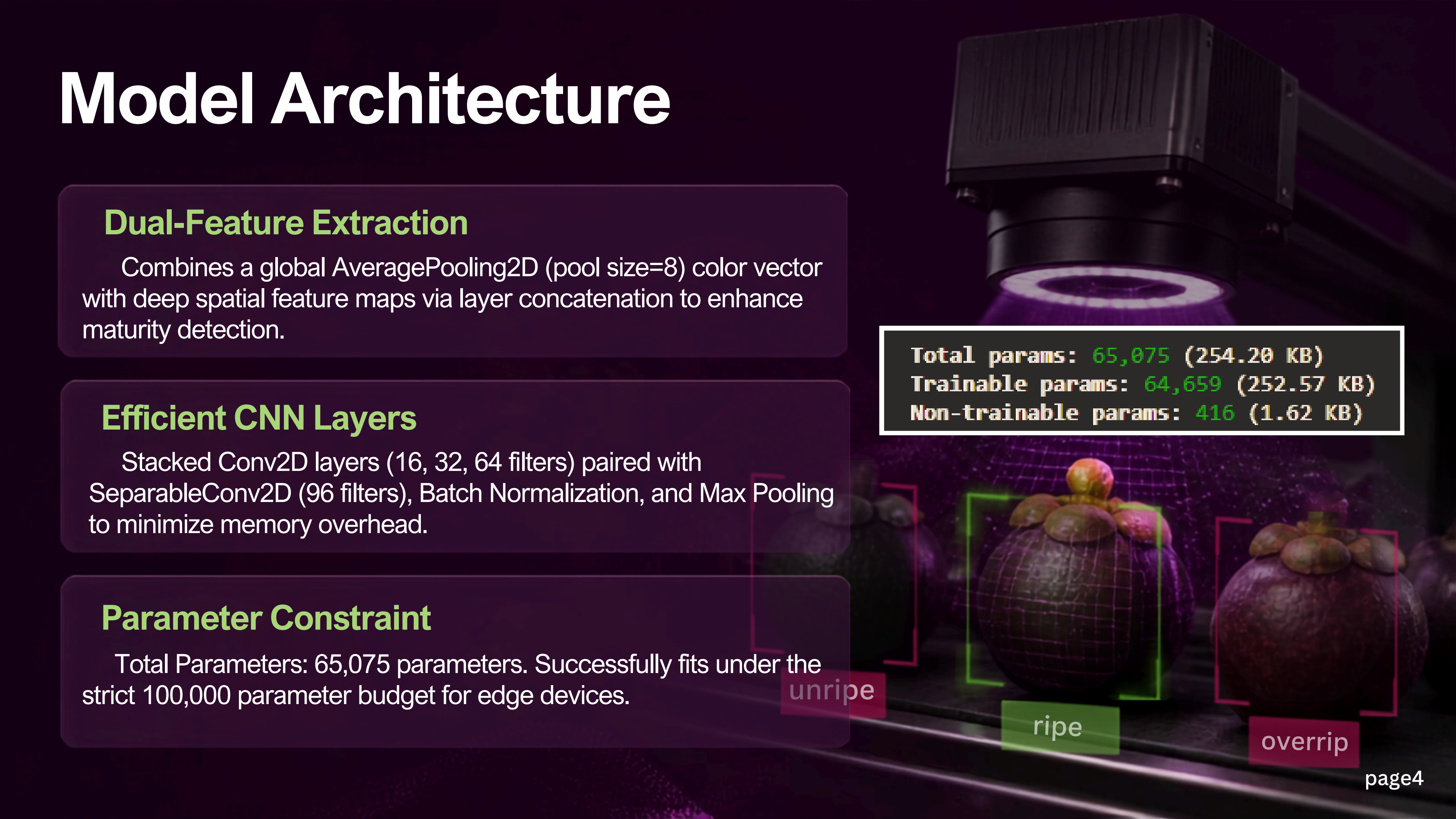
Combines a global AveragePooling2D (pool size=8) color vector with deep spatial feature maps via layer concatenation to enhance maturity detection.

Efficient CNN Layers

Stacked Conv2D layers (16, 32, 64 filters) paired with SeparableConv2D (96 filters), Batch Normalization, and Max Pooling to minimize memory overhead.

Parameter Constraint

Total Parameters: 65,075 parameters. Successfully fits under the strict 100,000 parameter budget for edge devices.



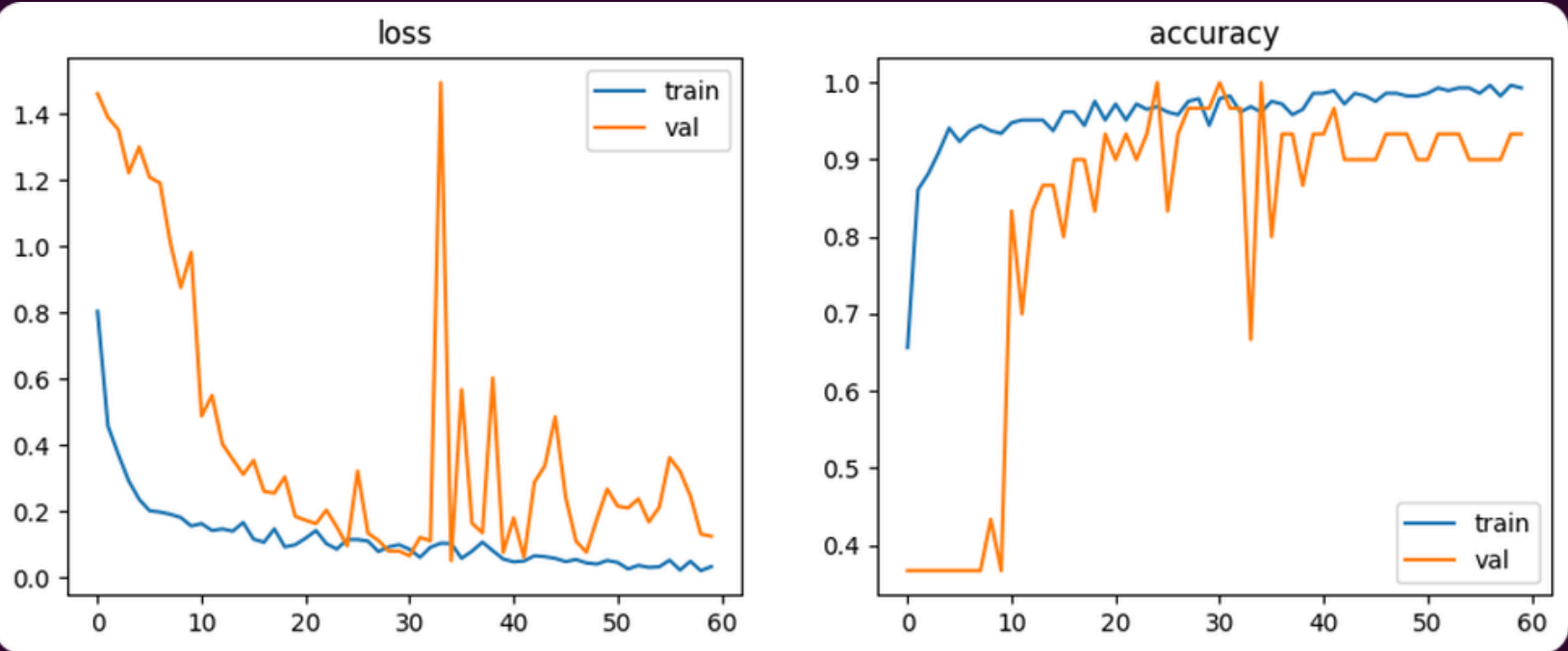
```
Total params: 65,075 (254.20 KB)
Trainable params: 64,659 (252.57 KB)
Non-trainable params: 416 (1.62 KB)
```

unripe

ripe

overrip

Quantization Results



Post-Training Quantization Summary

- Sanity Check: 100.00% accuracy verified on small sample batch during development.
- Validation Accuracy (Float32): 100.00% (Balanced Accuracy: 1.00).
- Test Accuracy (Float32 Baseline): 97.22% (1 Overripe sample misclassified as Ripe).
- INT8 quantization improved accuracy by 2.78 percentage points. (Full INT8 quantization improved test performance from 97.22% to 100.00%).

Metric	Float32	INT8
Validation accuracy	100.00%	—
Test accuracy	97.22%	100.00%
Balanced test accuracy	91.67%	100.00%
Accuracy change	Baseline	+2.78 pp

Hardware Setup

Edge & System Specification

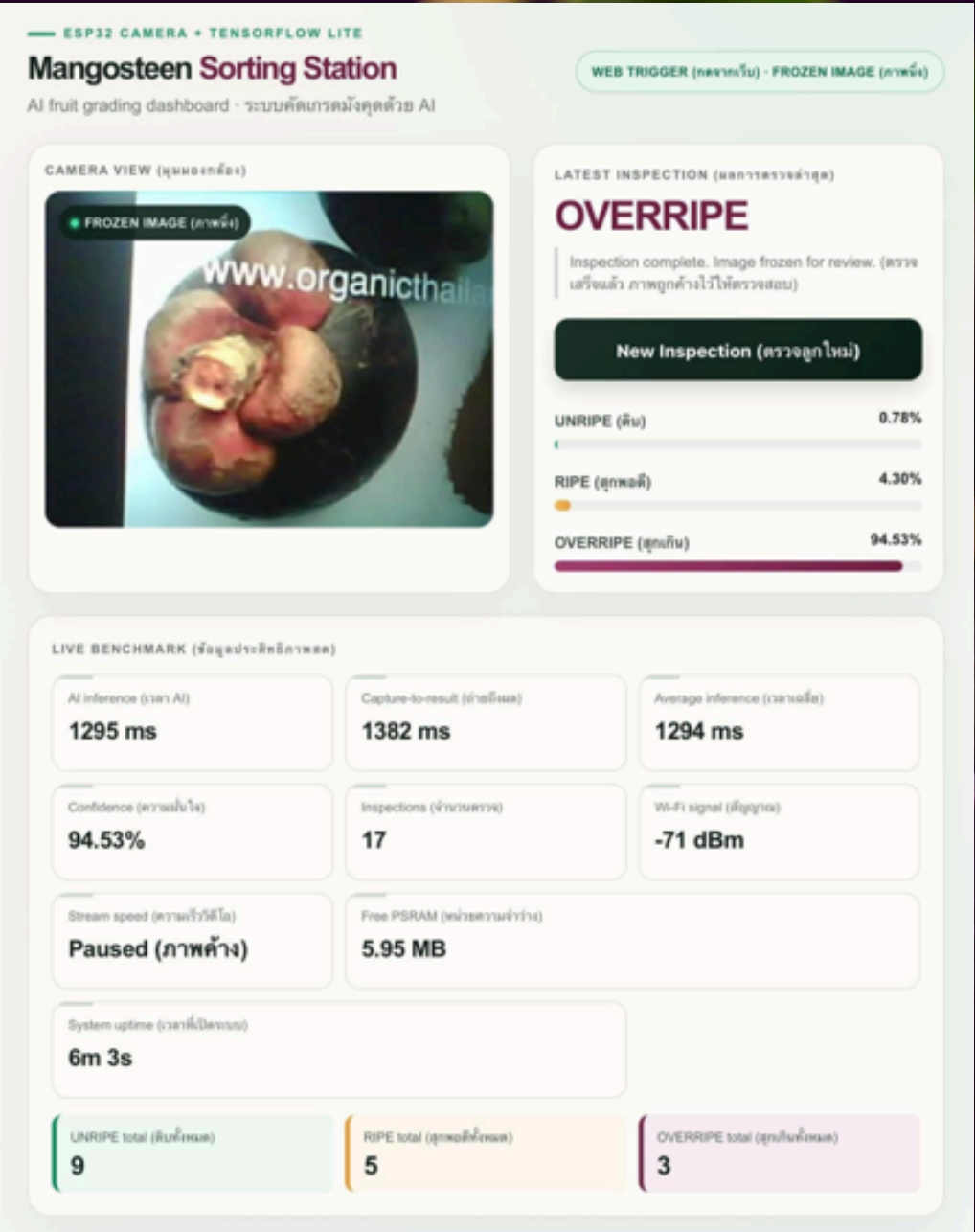
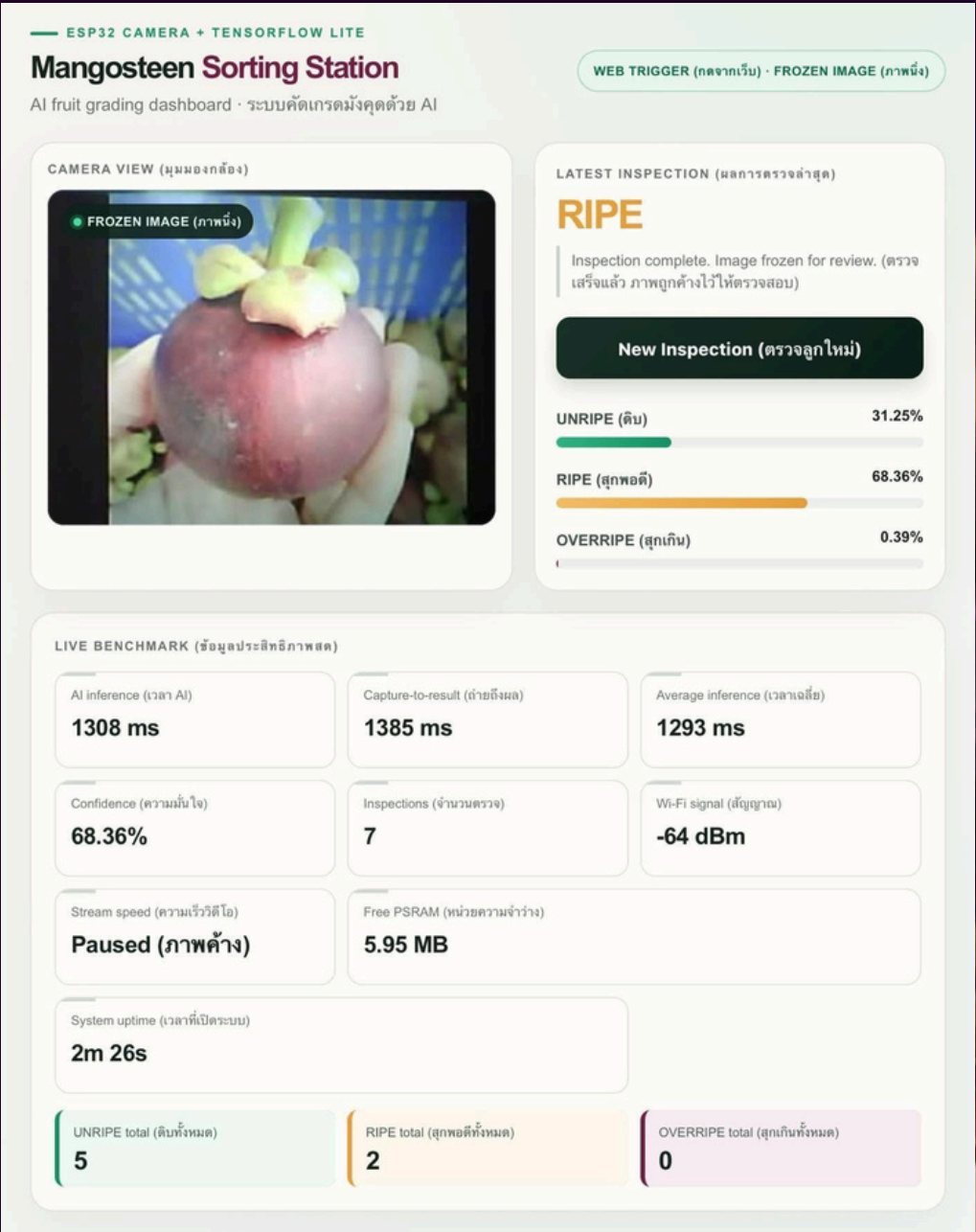
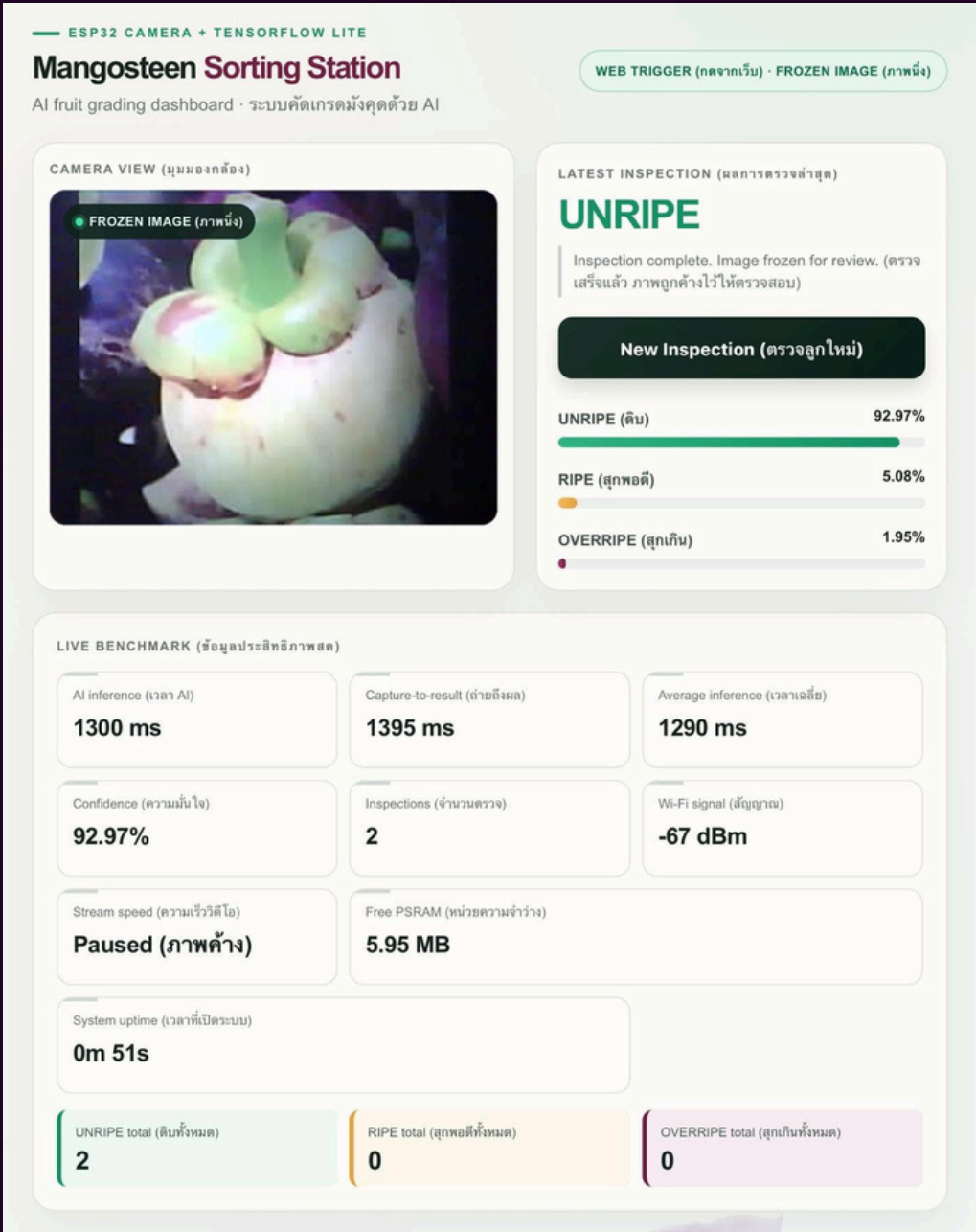
- Edge Controller:ESP32-S3-CAM running TensorFlow Lite Micro runtime.
- Available PSRAM: approx 5.95 MB Free Memory (sufficient for model tensor arena).
- Web Station Interface: Interactive "Mangosteen Sorting Station" Web GUI.
- Control Modes: Live Video Stream & Frozen Frame snapshot triggers via HTTP Web API.
- Accumulated Counters: Real-time tracking of total Unripe, Ripe, and Overripe counts.



Preliminary Benchmark Card

Metric	Resulting value
Colab Float32 accuracy	97.22%
Colab INT8 accuracy	100.00%
AI inference	1,281 ms
Capture-to-result	1,577 ms
Throughput แบบครบตั้งแต่ถ่ายถึงผล	0.63 images/s
Free PSRAM	5.95 MB

Live Demo & Key Failure Analysis



The ESP32-S3 successfully captured and classified live images of unripe, ripe, and overripe mangosteens.

Live Demo & Key Failure Analysis

Live Demo Evaluation

Web trigger correctly initiates single-frame capture and classifies fruit with high probability scores .

Dashboard UI seamlessly freezes captured frame and increments cumulative class counters.

Key Failure & Bottlenecks

Latency Limitation: The approximately 1.4-second end-to-end latency is acceptable for manual inspection but too slow for high-speed industrial conveyor sorting.

Weak Wi-Fi may increase dashboard response time, but its effect was not benchmarked separately.



Standardized Comparison Table

Metric	Our group	Other group	Difference
Test accuracy Float32	97.22%	75.0%	+22.22 percentage points
Test accuracy INT8	100%	58.30%	+41.70 percentage points
AI inference time	1,281 ms	~1,200 ms	≈+81 ms
Capture-to-result	1,577 ms	~1,500 ms	≈+77 ms
Free PSRAM	~5.95 MB	~5.90 MB	+0.05 MB
Throughput	0.63 FPS	0.67 FPS	-0.04 FPS (Other groups are faster)

Conclusions & Future Work



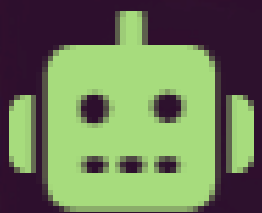
Edge AI Feasibility

100% accuracy is from Colab. The ESP32-S3 live demo worked, but full on-device testing is pending.



Latency Optimization

Future work will focus on optimizing TFLite micro kernels and camera frame capture buffers to lower inference time below 500 ms.



Hardware Actuation

Integrate ESP32 GPIO pins with mechanical pneumatic actuators to build a complete physical sorting mechanism for smart agriculture.